

First-Class Refinement Types

[Matt Bovel](#), EPFL

co-supervised by Viktor Kunčák ([Stainless](#)) and Martin Odersky ([Scala](#)) January 17, 2026

work done in collaboration with Quentin Bernet and Valentin Schneeberger

for Scala 3

Refinement types

Refinement types are types qualified with logical predicates.

For example, $\{ x: \text{Int} \mid x > 0 \}$ denotes the type of all integers x such that $x > 0$.

Implemented in many languages: [Liquid Haskell](#), [Boolean refinement types in F*](#), [Subset types in Dafny](#), etc.

Prior art in Scala: [SMT-based checking of predicate-qualified types for Scala](#) (Schmid and Kunčák, Scala Symposium 2016), [Refined](#), [Iron](#).

Outline